



College of

EDUCATION

AN INTRODUCTION TO MEASUREMENT MODELS AND STRUCTURAL EQUATION MODELING

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- A simulated data example to provide context
- Definitions of measurement models
- Concepts from construct measurement methods
- What is structural equation modeling and why it is used

- To introduce and motivate SEM, and to review some prerequisites, we will make use of an example data set
- Data come from a (simulated) sample of 150 participants who provided self-reports of a happiness scale and their marital status
- Participant responded one survey:
 - 5-item happiness survey (each item used roughly a 5-point Likert scale)
 - 1-item marital status question (are you married? Yes/No)
- The researchers were interested in the effects of marital status on happiness

PREDICTING HAPPINESS FROM THE GENERAL LINEAR MODEL

- We will now build a linear model for predicting a person's happiness, using marital status as the predictor
- We will use the total score of the 5-item happiness scale as our dependent variable (happiness_sumscore)
- We will code marital status with a 1 if a person is married and a 0 if they are not married
- Our beginning model is that of an **empty model** – no predictors for happiness (an **unconditional model**)
- Our ending model adds the marital status predictor

- Linear model: $Y_p = \beta_0 + e_p$
where $e_p \sim N(0, \sigma_e^2)$

```
#model 1 syntax
model01.syntax = "

# regression model
happiness_sumscore ~ 1 + married

# residual variance
happiness_sumscore ~~ happiness_sumscore

"
```

- Estimated Parameters (from lavaan output page):

Intercepts:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
➤ $\beta_0 = 14.740 (0.211)$	happinss_smscr 14.740	0.211	69.732	0.000	14.740	5.694

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
➤ $\sigma_e^2 = 6.702 (0.774)$	happinss_smscr 6.702	0.774	8.660	0.000	6.702	1.000

- Linear model: $Y_p = \beta_0 + \beta_1 \text{Married}_p + e_p$
where $e_p \sim N(0, \sigma_e^2)$



➤ Estimated Parameters

- $\beta_0 = 1.077$ (0.415; $R^2 = 0.043$); $p = 0.009$
- $\beta_1 = 14.245$ (0.281)
- $\sigma_e^2 = 6.414$ (0.741)

R-Square:

	Estimate
happinss_smscr	0.043

➤ Conclusion:

- Those married had significantly higher happiness scores than those unmarried

#model 1 syntax

model01.syntax = "

regression model

happiness_sumscore ~ 1 + married

residual variance

happiness_sumscore ~~ happiness_sumscore

"

Regressions:

	Estimate	Std.Err	z-value	P(> z)
happiness_sumscore ~				
married	1.077	0.415	2.595	0.009

Intercepts:

	Estimate	Std.Err	z-value	P(> z)
.happinss_smscr	14.245	0.281	50.620	0.000

Variances:

	Estimate	Std.Err	z-value	P(> z)
.happinss_smscr	6.414	0.741	8.660	0.000

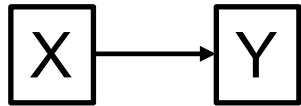
REVIEW OF MEASUREMENT (RELIABILITY)

- The general linear model (regression/ANOVA combo) and all models to this point in the course assumed that all variables were measured without error
- Surveys, questionnaires, and tests are all psychological instruments that are imprecise
 - Even adding up scores is a very naïve statistical model (called the parallel items model)
 - So, summed scores likely all have measurement error
- Related to the measurement error of a test score is the reliability of a test
 - Reliability is the amount of precision in a test (higher precision means less error)
 - The GLM assumes reliability = 1 for all variables
- In our example so far, our happiness variable was the sum score for a 5-item happiness scale

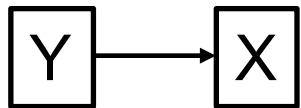
- “Test theory” is an abbreviated expression for:
 - “Theory of Psychological Tests and Measurements”
 - Or “Psychometric Theory” (even when not used in Psychology)
- Test theory is a general collection of statistical models for evaluating the development and use of instruments
 - Operationalize practical problems in measurement
 - Provide answers to practical problems in measurement
 - So yes, measurement models are indeed statistical models!
- There are (at least) three prominent branches of measurement models for latent traits
 - Classical test theory (CTT)
 - (Confirmatory) factor analysis (CFA)
 - Item response theory (IRT)
- All branches are heavily related but often are not taught as such

- Recall from introductory statistics that if two variables x and y are correlated, there are at least three reasons why:

1. Variable x causes variable y

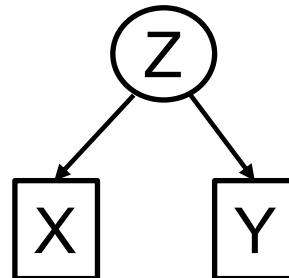


2. Variable y causes variable x



3. A third (or possibly many other) variable z causes x and y

- No direct relation between x and y



- Measurement models seek to determine z

- What you first learned about measurement probably falls under the category of Classical Test Theory (CTT):
 - Writing items and building scales (or “tests”)
 - Item analysis for differentiating “good” from “bad” items
 - Evaluating dimensionality underlying the items
 - Interpretating scale or test “scores”
 - Evaluating reliability and construct validity

- Big picture: CTT is a model with a restrictive set of assumptions within a more general family of latent trait measurement models

- **Latent trait** = Unobservable construct (“factor”)
 - Many types of variables: ability, attitude, tendency, etc.
 - e.g., “Intelligence”, “Extroversion”, “Depression”
- But how can we measure something unobservable?
 - Build **measurement models** by which to represent them!
- Big picture: Latent traits can be measured using observed responses to “**items**”/“**indicators**”
 - A new latent variable is created from the common variance across these observed responses posited to measure the same construct

- What do we call the latent trait measured by the indicators?
 - Classical Test Theory (CTT) → “True Score” (T)
 - Confirmatory Factor Analysis (CFA) → “Factor Score” (F)
 - Item Factor Analysis (IFA) → “Factor Score” (F)
 - Item Response Theory (IRT) → “Theta” (θ)

- Fundamental difference in approach:
 - CTT → unit of analysis is the WHOLE TEST (item sum or mean)
 - Sum = latent trait, so items and persons are inherently tied together → bad
 - Only using the sum requires restrictive assumptions about the items
 - CFA, IFA, IRT, and other measurement models → unit of analysis is the ITEM
 - Model of how item response relates to a separately estimated latent trait
 - Provides way of separating item and person properties → good for flexibility
 - Different names of models are used for differing item response formats
 - Provides a framework for testing adequacy of measurement models

- Families of measurement models are labeled differently based on their indicators' response format:
 - Continuous responses? → Confirmatory Factor Models
 - Categorical responses? → Item Response Theory or Item Factor Models
 - Measurement models for other response types exist too (like counts), but they don't necessarily have special names (I say "generalized")

- Other relevant, related terms:
 - Structural Equation Modeling (SEM) is regression within and between the latent variables defined by the measurement models and other observed variables
 - Sometimes called Path Analysis with Latent Variables
 - Path Analysis is used to describe multivariate regression among observed variables only
 - Concepts such as mediation (indirect effects) and moderation (interactions between variables) carry over into SEM
 - Although interactions between latent and observed variables become tricky to estimate

- Motivated by problems in education and psychology
 - Education → Assessment of academic abilities
 - Psychology → Understand structure of intelligence or personality
 - Piecemeal approach; also barriers from technical presentation
 - Theories developed before availability of computing power, so approximations were developed that could actually be used (with remnants that unfortunately still get used, like alpha and EFA)

- 1904: Charles Spearman published two seminal papers
 - One showed how to estimate amount of error in test scores
 - Led to classical true score theory (aka, classical test theory)
 - Other showed how to recognize from test data that the tests measure just one psychological attribute in common (“G”)
 - Led to common factor theory (aka, confirmatory factor analysis)

- In CTT, the test is the unit of analysis:

$$Y_{total} = T + e$$

- **True score T:** best estimate of construct
 - **Error e:** mean of zero, uncorrelated with T
-
- Variance of test scores: $\sigma_Y^2 = \sigma_T^2 + \sigma_e^2$
-
- Reliability is defined as the proportion of test variance accounted for by true score variance:

$$\rho = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_e^2}$$

- Items assumed exchangeable (they all count the same)
 - More items means higher reliability, regardless of type of item or quality of item

- CTT unit of analysis is the TEST or SCALE (sum/mean of items)
 - Want to quantify how much of observed test score variance is due to true score variance versus error variance
 - Error is a unitary construct in CTT (and error is always bad)
 - Goal is then to reduce “error” variance as much as possible
 - Standardization of testing conditions (make confounds constants)
 - Aggregation → more items are better (errors should cancel out)
 - Items are exchangeable; item properties are NOT taken into account in indicating the latent trait of a given person (which is just the sum)
- Followed by *generalizability theory* to distinguish kinds of error
 - e.g., item variance, person variance, rater variance, occasion variance
 - Modern analog: mixed-effects (multilevel) models with crossed random effects for each (random) sampling dimension and their interactions

- Brief history of solutions for quantifying reliability:
 - 1904: Spearman: from alternate forms or test-retest
 - 1945: Guttman: from the relations between the items within a test (i.e., coefficient alpha)
 - 1951: Cronbach further developed Guttman's work → "Cronbach's alpha"
 - Called "Guttman-Cronbach alpha" by McDonald (and no one else)
 - Cronbach's work further elaborated into generalizability theory
 - And no, a good alpha doesn't mean anything—stay tuned for why!
 - 1950: Gulliksen classic text for CTT
 - See also Nunnally's texts from the 1970's–1990's

- The happiness sum score was just the sum of the five happiness items in our data set for each person:

$$Y_p = X_1 + X_2 + X_3 + X_4 + X_5 = \sum_{i=1}^I X_i$$

- The sample correlation matrix of these data are:

```
> cor(data02[,3:7])
```

	X1	X2	X3	X4	X5
X1	1.00000000	-0.06354391	0.06019113	0.02107528	0.15239670
X2	-0.06354391	1.00000000	0.07555098	-0.02271474	0.03080322
X3	0.06019113	0.07555098	1.00000000	0.05662233	0.15168240
X4	0.02107528	-0.02271474	0.05662233	1.00000000	0.16500076
X5	0.15239670	0.03080322	0.15168240	0.16500076	1.00000000

- For quantitative items (items with a scale – although used on categorical items), this is often indexed by Cronbach's Alpha...
 - Or 'Guttman-Cronbach alpha' (Guttman 1945 > Cronbach 1951)
 - Another reduced form of alpha for binary items: KR 20
- Alpha is described in multiple ways:
 - Is the mean of all possible split-half correlations
 - Is expected correlation with hypothetical alternative form of the same length
 - Is lower-bound estimate of reliability under assumption that all items are tau-equivalent (more about that later)
 - As an index of "internal consistency"
- Note: Alpha is used to describe the reliability of a sum score assuming a tau-equivalent model
 - Tau-equivalent model: One where variances differ but correlations are the same among items
- Alpha, however, is calculated using covariance matrices instead of correlation matrices

- First, we need the covariance matrix of all the items

```
> cov(data02[,3:7])
```

	X1	X2	X3	X4	X5
X1	1.19557420	-0.07495136	0.06737788	0.02346632	0.16606757
X2	-0.07495136	1.16368298	0.08343611	-0.02495217	0.03311574
X3	0.06737788	0.08343611	1.04807695	0.05902929	0.15475788
X4	0.02346632	-0.02495217	0.05902929	1.03697078	0.16745194
X5	0.16606757	0.03311574	0.15475788	0.16745194	0.99321203

- The **sum of the item variances** is given by:

$$ItemVar(Y) = \sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2 + \sigma_5^2 = 5.438$$

- The **variance of the sum of the items** is given by the sum of ALL the item variances and covariances:

$$TotalVar(Y) = \sum_{i=1}^I \sum_{j=1}^J \sigma_{ij} = 6.747$$

$$\alpha_{GC} = \frac{I}{I - 1} \cdot \frac{TotalVar(Y) - ItemVar(Y)}{TotalVar(Y)}$$
$$\alpha_{GC} = \frac{5}{5 - 1} \cdot \frac{6.747 - 5.437}{6.747} = 0.243$$

- Numerator reduces to just the covariance among items
 - Sum of the item variances...
 - $Var(X) + Var(Y) = Var(X) + Var(Y) \rightarrow$ just the item variances
 - Variance of total Y (the sum of the items)...
 - $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y) \rightarrow$ PLUS covariances
 - So, if the items are related to each other, the variance of the total Y item sum should be bigger than the sum of the item variances
 - How much bigger depends on how much covariance among the items – the primary index of relationship
- **Therefore – the happiness scale is not perfectly reliable**
 - Violates the assumption of the GLM/Regression

- Main idea: Build a measurement model of which response indicators should measure the same thing
 - **CFA = Linear regression model** predicting each continuous observed outcomes (items/indicators) from **latent** trait (unobserved) predictor(s)
- Differs from exploratory factor analysis (that is often NOT a model):
 - In CFA *you* impose the number and content of factors
 - In CFA alternative models are COMPARABLE and TESTABLE
- Uses of confirmatory factor analysis models:
 - Analyze relationships among indicators that have normal, continuous distributions
 - Note: Likert items use IRT (but many features of CFA carry over)
 - Provide separation of persons, items, and occasions (as in any measurement model)

- **The CFA unit of analysis is the ITEM (as in any measurement model):**

$$y_{ip} = \mu_i + \lambda_i F_p + e_{ip} \rightarrow \text{both items AND people matter}$$

$$y_{ip} = \beta_{0i} + \beta_{1i}x_p + e_{ip}$$

- Observed response for item i and person p
 - = intercept of item i (μ)
 - + person p 's latent trait/factor (F), item-weighted by factor loading of item (λ)
 - + error (e) of item i and person p
- **Dimensionality** → part of the model
 - Local Independence → e residuals are independent after controlling for factor(s)
 - The factor is the reason why item responses were correlated in the first place!
 - If not, you can **augment the model** to address unintended multidimensionality
- **Linear model** → a one-unit change in latent trait/factor F_p creates same increase in the expected response y_{ip} along all points of y_{ip}
- Items can now differ from each other in how much they relate to the latent trait, *but a “good item” is assumed equally good for everybody!*

- 1900's: Spearman's "G" single-factor models
 - Development of techniques designed to find a common factor
 - Led to development of other IQ tests (Stanford-Binet, Wechsler)

- 1930's and 1940's: Thurstone elaborated Spearman's "G" unidimensional model into a "multiple factor" model
 - Beginnings of exploratory factor analysis to do so
 - Later applied in other personality tests (e.g., MMPI)

- 1940's and 1950's: Guttman's work
 - Factor analysis and test development is about generalizing from measures we have created to more measures of the same kind
 - Thus, need to think about measurement structure before-hand

- 1940s: Lawley → rigorous foundation for statistical treatment of common factor analysis
 - But had to wait for better computers to be able to do it!
- 1952: Lawley → beginnings of confirmatory factor model
 - Later extended by Howe and Bargmann (1950's)
 - Further extended by Jöreskog
- But this linear model *pry* should not be applied to binary, ordinal, or other not-continuous responses...
 - Predicted response will go past possible response options
 - Errors can't be normally distributed with constant variance
- So then what? Item Response Theory to the rescue...
 - *aka*, measurement models for generalized response formats

- IRT resulted from combination of ideas from factor analysis and phi-gamma law of psychophysics
 - When detecting stimuli of varying intensity (e.g., light), the response follows a smooth, S-shaped curve that can be represented by the cumulative normal distribution
 - That response function also works to model probability of a correct response given (1 to 4) model parameters
- 1950: Lazarsfeld: Introduced “latent structure analysis” → factor analysis for binary item responses
 - Beginnings of item response theory (which is not a theory per se, but another set of latent trait measurement models—latent class/profile models)

- Linear regression is to confirmatory factor models as to:
 - Logistic regression is to binary IRT models
 - Ordinal/nominal regression is to polytomous (more than two category items) IRT models
 - IRT = generalized linear model predicting each categorical observed outcome indicator from latent predictors using link functions
 - Term “IRT” usually goes with full-information estimation (use all data)

- A Rasch model is a restricted version of an IRT model

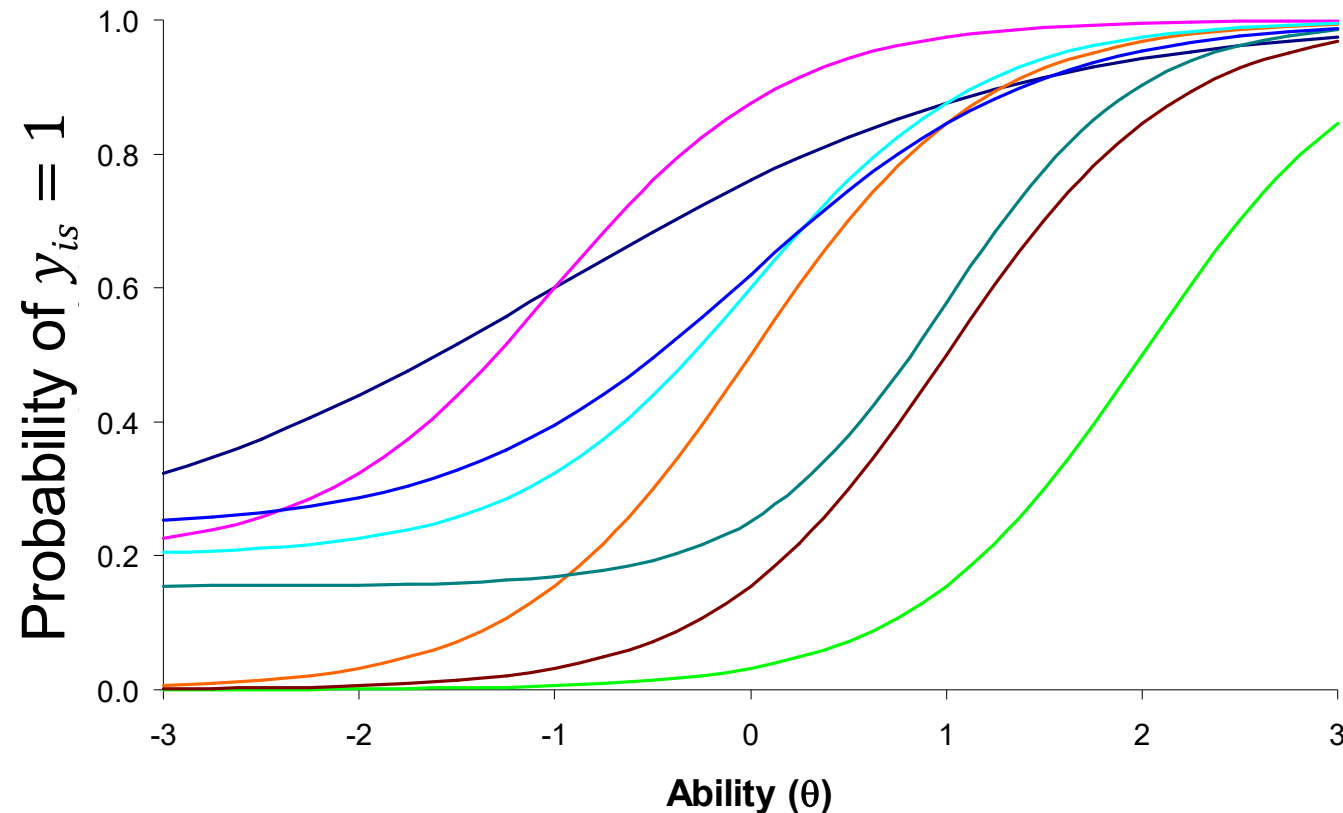
- Uses of IRT models:
 - *Correctly* analyze categorical indicators (binary, ordinal, or nominal)
 - Examine sensitivity of measurement across range of latent trait
 - Provide separation of persons, items, and occasions (as in any measurement model)

a = Discrimination = slope of ‘line’

b = Difficulty = location of ‘line’

c = Lower Asymptote of ‘line’

d = Upper Asymptote of ‘line’



Note: Theta ability has a nonlinear relationship to probability of item response, but a linear relationship to its link-transformed mean...

- The **IRT** unit of analysis is the individual **ITEM** (as in any measurement model)
Link(y_{ip}) = $a_i(\theta_p - b_i) \rightarrow$ both items AND subjects matter
 - Response probability is predicted via a link (transformation) function (usually logit or probit, in which probit is called “ogive” in IRT)
 - Items and persons are located on the same latent metric
 - Probability of getting an item right depends (at least) on the subject’s ability (θ_p = “**Theta**”) and the item’s difficulty (b_i), weighted by its discrimination (a_i , how related the item is to the latent trait)
 - “**Item factor analysis**” (**IFA**) re-arranges IRT model into something that looks more like CFA (and usually uses limited information estimation)
- All items are NOT created equal (not exchangeable)
 - Having items that differ in their properties is a GOOD THING, because you can customize tests for different groups or purposes
 - Reliability (“information”) varies across trait level, and depends specifically on how well the items’ difficulty matches subjects’ traits

- 1952: Lord: Spearman's single-factor model can be applied to dichotomous items
 - Binary responses modeled by normal ogive function ("probit")
 - Later work used easier logit link instead ($\text{logit} \approx \text{probit} \times 1.7$)
 - Extended in 1960s by Birnbaum (and others)
- 1968: Lord & Novick → first CTT text to also include IRT
 - Well-connected to emerging scholars in both educational testing and psychometric methods
- 1960: Separate work by Rasch (common 'a' parameter)
 - Restricted IRT model, but with desirable properties if it fits...
 - ... and a very different philosophical viewpoint (as "the" model)

- Classical test theory can be viewed as a restricted form of the common factor model, but the focus is the TEST...
 - Originated by Spearman, elaborated by Thurstone, formalized by Lawley, and made practical in software by Jöreskog
- Item response theory (and Rasch) models are common factor models used for binary or ordinal responses...
 - Developed by Lord, Birnbaum, Rasch, and their students
- Confirmatory factor analysis are common factor models for continuous responses...
 - Approximation for ordinal data with varying degrees of success
- Latent traits can also be indicated by other kinds of non-normal responses (count, zero-inflated, two-part/hurdle)....
 - But they don't have special names (I'd call it "generalized SEM")
 - Other response data (e.g., eye fixation, RT) can be used, too!

- Explicit, testable models of dimensionality
- Concrete guidelines for selecting items to build scales
- Assess measurement sensitivity across range of latent trait (i.e., know where the “holes” of imprecision are)
- Provide comparability across persons, items (different forms scales or different scales), and occasions
- Examine comparability across groups or repeated measures
 - Confirmatory factor analysis → “Measurement invariance”
 - Item response theory → “Differential item functioning”
- Internal and external evidence for construct validity
- Generalized measurement models can even accommodate different response formats within the same instrument

- Primary: Required sample size
 - Sample sizes of 100s for sure, and preferably 1000s
 - Bayesian methods are much more flexible!
 - Uses maximum likelihood (limited-info WLSMV estimator in Mplus can also be used for multidimensional IRT models)
- Technical difficulties
 - Estimation is harder, especially in multidimensional IRT
 - References written in Greek (literally)
- Misnomers about what measurement models (within SEM) can do...
 - Bad items are still bad items, no matter what model is used
 - No, SEM is still not causal modeling

A CONFIRMATORY FACTOR ANALYSIS OF OUR FIVE-ITEM HAPPINESS SCALE

- The CFA model is essentially a set of simultaneous regressions:

$$X_{p1} = \mu_1 + \lambda_1 HPP_p + e_{p1}$$

$$X_{p2} = \mu_2 + \lambda_2 HPP_p + e_{p2}$$

$$X_{p3} = \mu_3 + \lambda_3 HPP_p + e_{p3}$$

$$X_{p4} = \mu_4 + \lambda_4 HPP_p + e_{p4}$$

$$X_{p5} = \mu_5 + \lambda_5 HPP_p + e_{p5}$$

- X_{pi} ($i \in \{1, \dots, 5\}$) is the response of person p to item i
- HPP_p is the score on the latent (unobserved) happiness variable
 - We have to assume we observed it to build this model
- μ_i is the intercept for item i
- λ_i is the factor loading of item i
 - As our model has only one latent variable, there can be only one loading per item—different for multiple latent variables (multidimensional models)
- e_{pi} is the residual for person p and item i

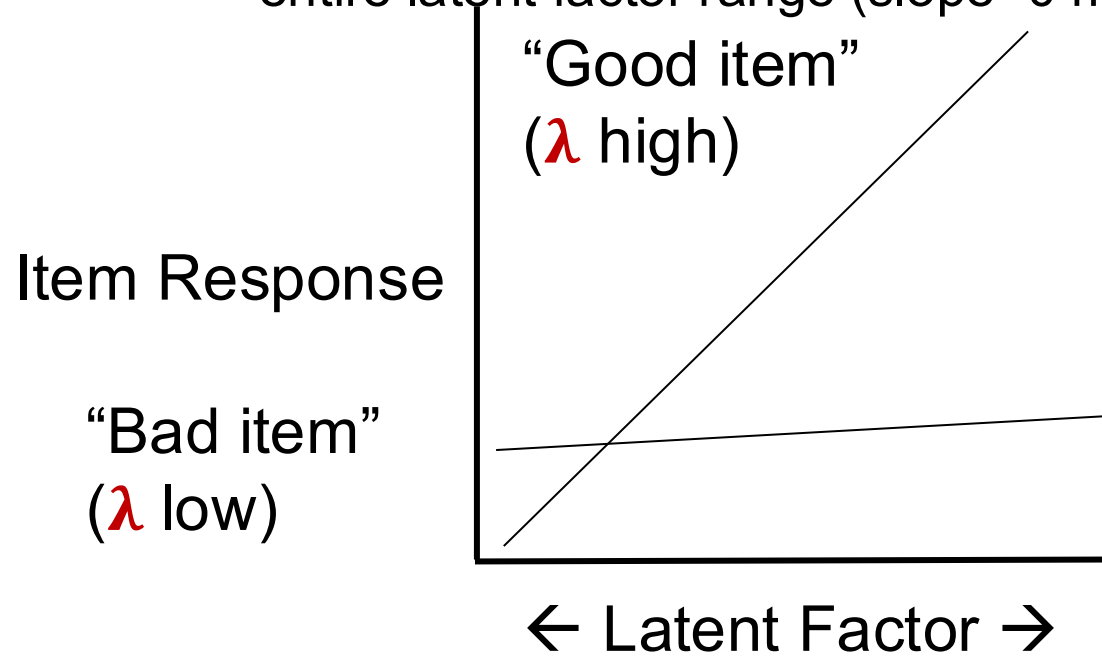
- Data going in to be recreated by the CFA model parameters
= item mean vector + item variance–covariance matrix
- CFA item intercepts (μ_i) try to recreate the item means
 - Item means are unconditional; item intercepts are conditional on $Fs = 0$
 - When each item gets its own intercept (the usual case), the item means should* be perfectly recreated (so no room for mis-fit or mis-prediction)
- CFA item error variances ($\sigma_{e_i}^2$) try to recreate the item variances
 - Item variances are unconditional; item error variances are conditional
(= leftover variance after accounting for the contribution of the factor)
 - When each item gets its own error variance (usual case), the item variances should* be perfectly recreated
(so no room for misfit or mis-prediction)
- CFA item factor loadings (λ_i) try to recreate the item covariances
 - Given 3+ items, there will be more covariances among items to predict than item factor loadings to recreate them, thus creating room for misfit

- **Item Discrimination:** How related each item is to the latent trait
 - In CTT, discrimination was given by the item–remainder correlation
 - The total score is the best estimate of the latent trait in CTT ($True_s \rightarrow F_s$ in CFA)
 - In **CFA**, discrimination is given by the **factor loading/slope** (λ_i)
 - We now have a separate factor F_s (whose job is to recreate the item covariances)
 - **Stronger factor loadings indicate better, more discriminating items**
 - Use the “standardized” factor loadings when comparing across items

- **Item Difficulty/Severity:** Location of item on the latent trait metric
 - In CTT, difficulty is given by the item mean – which is sample-dependent
 - In **CFA**, difficulty is given by the **item intercept** (μ_i) – which is sample-free
 - In contrast to other latent trait models (such as IFA/IRT), difficulty (or “easiness” because we are using additive intercepts) are often ignored in CFA... here’s why...

A “**good**” item has a **steep** slope (i.e., factor loading) in predicting the item response from the factor. Because this is a **linear slope**, the item is assumed to be equally discriminating (**equally good**) across the entire latent factor.

Similarly, a “**bad**” item has a **flatter** slope that is **equally bad** across the entire latent factor range (slope=0 means item is unrelated to factor).

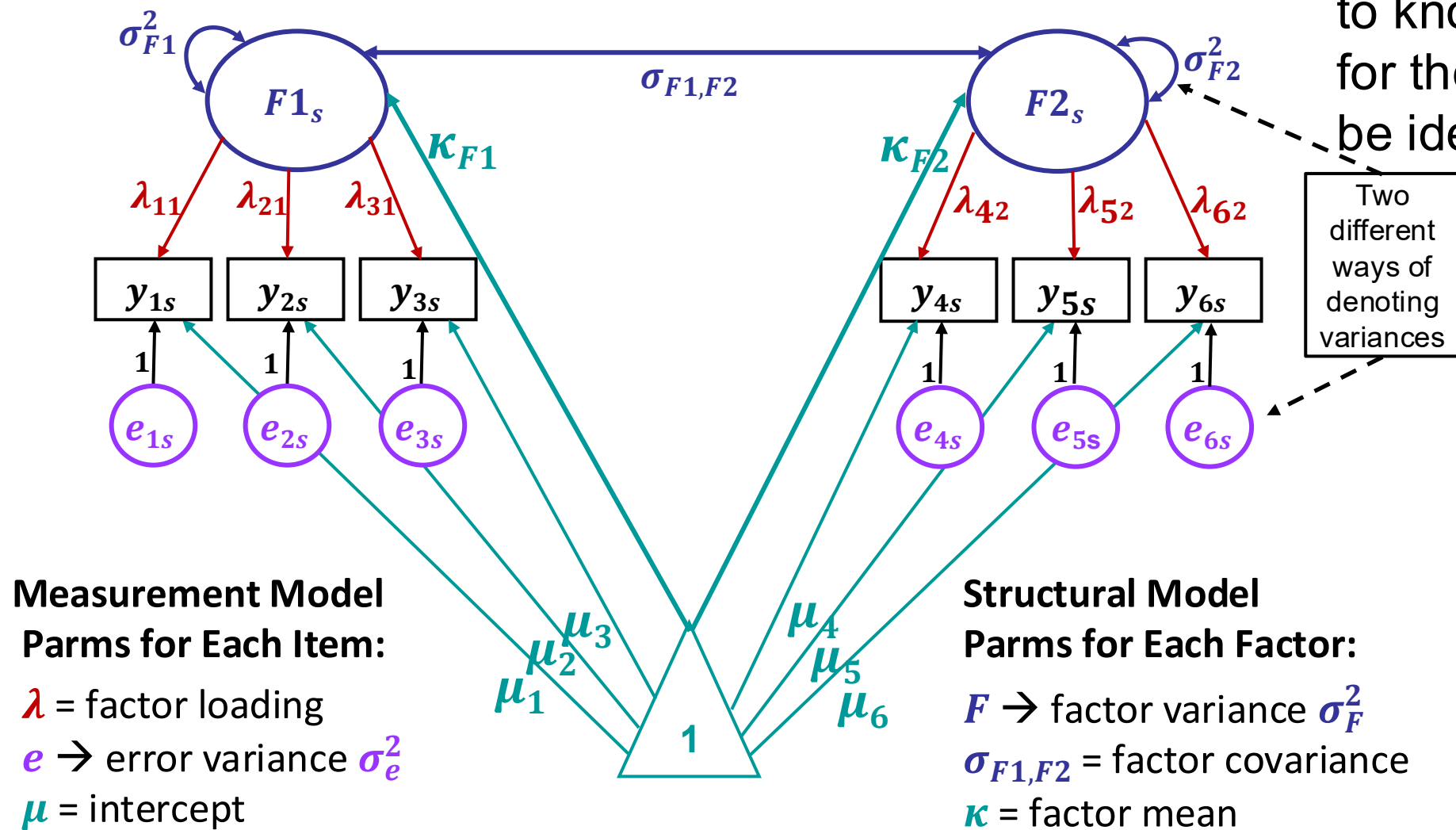


Here item intercepts are irrelevant in evaluating how “good” an item is, so they are not really needed in CFA.

But we will estimate item intercepts, because they are critical when:

- Testing factor mean differences in any latent factor model
- Items need to have a nonlinear slope in predicting the item response from the factor (IRT)

Example Diagram of Two-Factor CFA Model



But some parameters will have to be **fixed** to known values for the model to be identified.

- We don't have observed F values (called “factor scores”) by which to predict the item responses, but we don't need them!
 - The F values are random effects assumed to have a multivariate normal (MVN) distribution, which means all we need to estimate the model are parameters for their means, variances, and covariances
- The role of the CFA model parameters is to recreate the summary statistics of the item (indicator) responses
 - In CFA, these are the item means, variances, and covariances (Pearson-style), which are sufficient statistics assuming a multivariate normal conditional distribution for the item responses from the same subject
 - In IRT and IFA, item summary statistics will vary by model and estimator (because variance is not separately estimated for categorical outcomes)
- How well the observed item summary statistics match those recreated by the CFA model is the primary basis of model fit...

- Unstandardized → predicts scale-sensitive original item response:
 - **Uses scale-specific regression model:** $y_{is} = \mu_i + \lambda_i F_s + e_{is}$
 - *Useful when comparing solutions across samples (when original values matter)*
 - Unstandardized model parameters predict **item means and variance–covariance matrix**
 - Note the solution asymmetry: item parameters μ_i and λ_i are given in the original metric, but e_{is} is replaced as the error variance across subjects for that item (squared metric)
 - $Var(y_i) = [\lambda_i^2 * Var(F_s)] + Var(e_{is}) \rightarrow \sigma_{y_i}^2 = [\lambda_i^2 * \sigma_F^2] + \sigma_{e_i}^2$
- Standardized → Rescaled so $M(y_i) = 0$; $Var(y_i) = Var(F_s) = 1$:
 - *Useful when comparing items within a model → relative values on same scale*
 - Standardized model predicts the **item correlation matrix (b/c item means = 0)**
 - Standardized intercept = $\mu_i / SD(y_{is}) \rightarrow$ not typically reported (and mostly unhelpful)
 - Standardized factor loading = $[\lambda_i * SD(F_s)] / SD(y_{is}) =$ **item correlation with factor**
 - Standardized factor loading² = item R^2 = proportion of item variance DUE to the factor
 - Standardized error variance = $1 - \text{stand } \lambda_i^2$ = prop. item variance due to *NOT* the factor

Items from same factor (room for misfit=bad recreation):

- Unstandardized solution: $\text{Covariance}(y_1, y_3) = \lambda_{11} * \text{Var}(F1) * \lambda_{31}$
- Standardized solution: $\text{Correlation}(y_1, y_3) = \lambda_{11} * (1) * \lambda_{31} \leftarrow \text{std loadings}$
- ONLY reason for correlation is their common factor (local independence, LI)

Items from different factors (room for misfit=bad recreation):

- Unstandardized: $\text{Covariance}(y_1, y_6) = \lambda_{11} * \text{Cov}(F1, F2) * \lambda_{62}$
- Standardized: $\text{Correlation}(y_1, y_6) = \lambda_{11} * \text{Cor}(F1, F2) * \lambda_{62} \leftarrow \text{std loadings}$
- ONLY reason for correlation is the correlation between factors (again, LI)

Variances are additive (and should be reproduced correctly):

- $\text{Var}(y_1) = (\lambda_{11}^2) * \text{Var}(F1) + \text{Var}(e_i) \rightarrow \text{note imbalance of } \lambda^2 \text{ and } e_i$
(e_i is not a model parameter – its variance across subjects is instead)
- Same equation applies to both unstandardized and standardized solutions

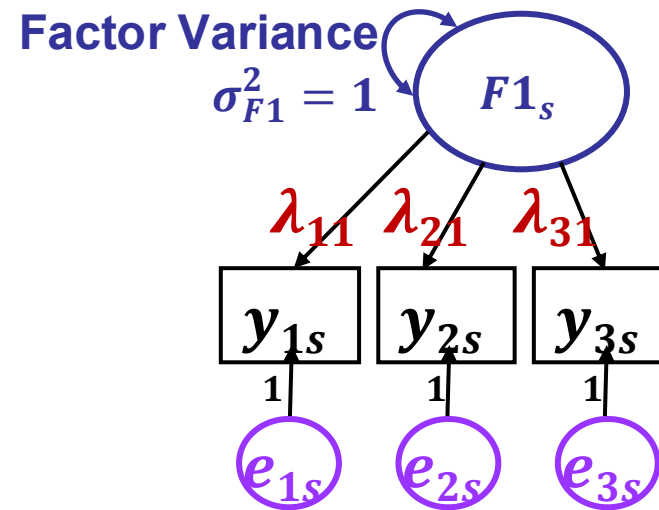
➤ **Part 1: Each latent factor has a scale**

- Each latent factor needs a mean and a variance
- Necessary but not sufficient for estimating the CFA model

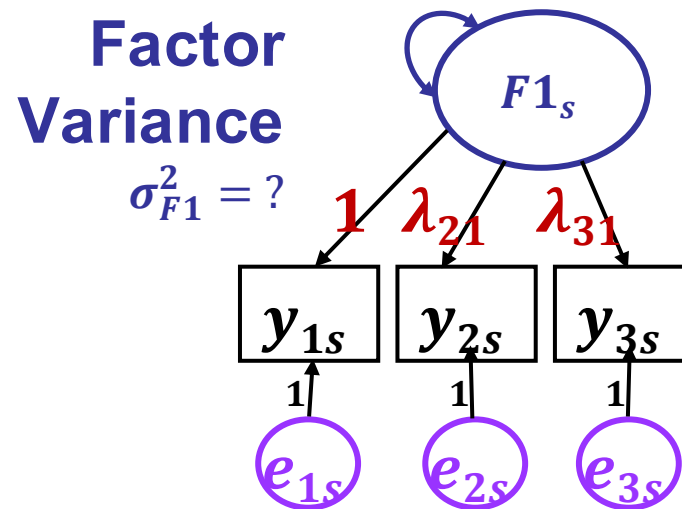
➤ **Part 2: Ensure the CFA model is estimable**

- Data going in versus estimated parameters going out:
 - Item **means** → item **intercepts** (usually 1:1 ratio)
 - Item **variances** → item **error variances** (usually 1:1 ratio)
 - Item **covariances** → item **factor loadings** (must have **ratio** ≥ 1)
- In practice, this means the number of estimated loadings may not exceed the number of observed item covariances
- This also means that each pair of variables has only one direct relationship (more of a concern in path models)

“Z-Score Factor” Method (1)



“Marker Item” Method (2)

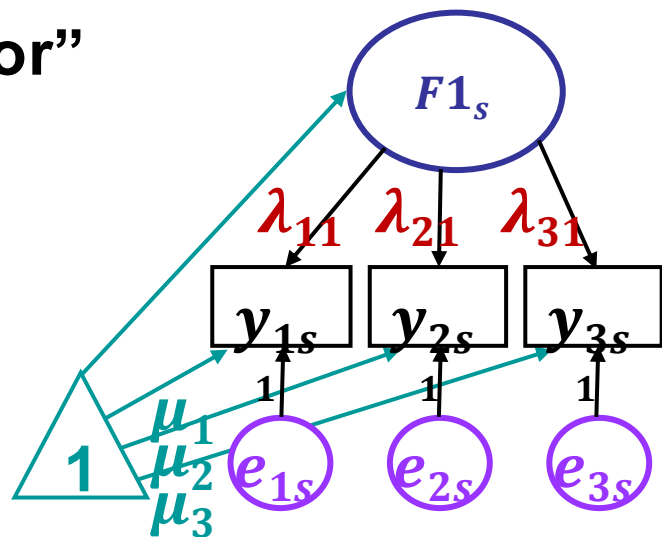


- The factor doesn't exist, so **it needs a scale** (it needs a mean and variance):
- There are two **equivalent** options to create a scale for the factor **VARIANCE**:
 - (1) **Fix factor variance to 1: “z-score”**
 - Factor SD is same as standard z-score (but any constant > 0 would also be ok)
 - **Shouldn't** be used in models with higher-order factors (coming later in this course)
 - (2) **Fix a “marker item” loading to 1**
 - Factor variance is then estimated the “reliable” part of the marker item variance
 - e.g., Std. loading = 0.9, item variance = 16?
Factor variance = $(0.9^2) * 16 = 12.96$
 - Can cause the model to blow up if marker item has no correlation with the factor at all

“Z-Score Factor” Method (1)

Factor Mean

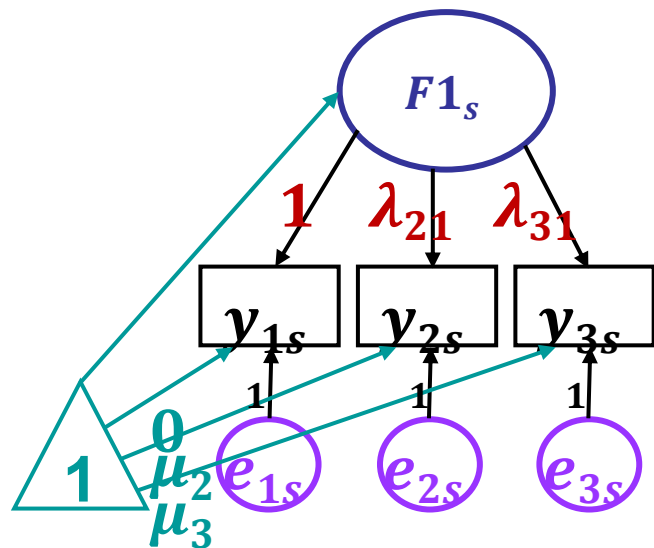
$$\kappa_1 = 0$$



“Marker Item” Method (2)

Factor Mean

$$\kappa_1 = ?$$



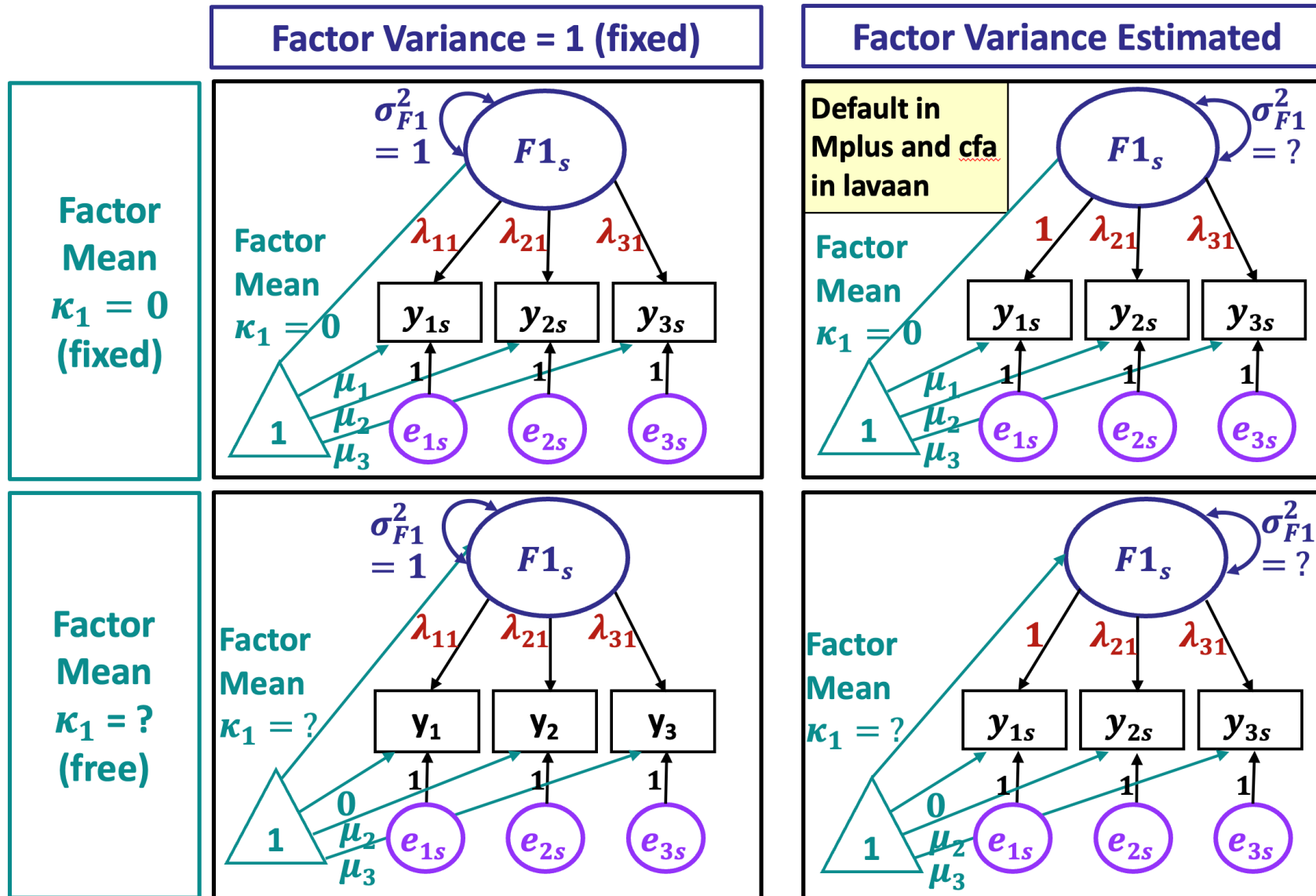
- The factor doesn't exist, so **it needs a scale** (it needs a mean and variance):
- There are two **equivalent** options to create a scale for the factor **MEAN**:

➤ (1) Fix factor mean to 0: “z-score”

- Factor mean is same as standard z-score (but any constant would also be ok)
- **Can** be used in models with higher-order factors (coming later in the course)
- Item intercepts = item means in this case

➤ (2) Fix a “marker item” intercept to 0

- Factor mean = mean of marker item
- Item intercepts = expected item responses when factor = 0 (→ marker = 0, which may not make sense for all item response scales), not often used (except growth models)

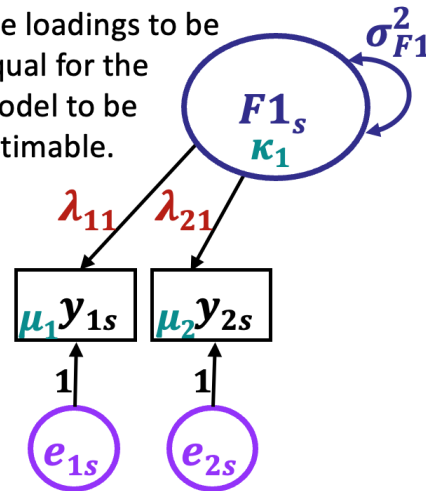


- After scaling the factors, then try to reproduce observed item means and variance–covariance matrix using as few estimated parameters as possible
 - (Robust) Maximum likelihood used to estimate model parameters
 - Measurement Model: Item intercepts, item factor loadings, item error variances
 - Structural Model: Factor means, factor variances, factor covariances
 - Global model fit = difference between model-recreated and data-observed variance–covariance matrix (but only covariances usually contribute to misfit)

- How many possible parameters can you estimate: What is the total DF?
 - Total **DF** = $\frac{v(v+1)}{2} + v$ where v is the # items (NOT people, like usual)
 - Total DF = number of item means, variances, and covariances
 - e.g., if $v = 4$ items, then Total DF = $\frac{4(4+1)}{2} + 4 = 14$
 - Model **DF** = data input – model output
 - Model **DF** = # possible parameters – # estimated parameters
 - Model **DF** = # parameter left over (could have been estimated, but didn't)

- Model is **under-identified** if there are more unknown parameters than item variances, covariances, and means with which to estimate them
- Model **cannot be estimated** because there are an infinite number of different parameter estimates that would result in the same (and perfect) fit
- Example: $x + y = 7$??

You'd have to constrain the loadings to be equal for the model to be estimable.



In other words, the assumptions required to calculate two-score reliability in CTT are the result of model under-identification.

Total possible **DF** = unique pieces of data = 5

0 factor variances

0 factor means

2 item loadings OR

2 item intercepts

2 error variances

1 factor variance

1 factor mean

1 item loading

1 item intercept

2 error variances

Model DF = 5 – 6 = – 1

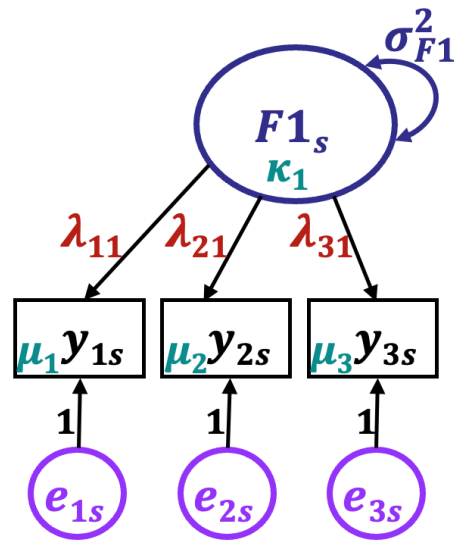
If $cor(y_1, y_2) = .64$, then:

$\lambda_{11} = .800, \lambda_{21} = .800$??

$\lambda_{11} = .900, \lambda_{21} = .711$??

$\lambda_{11} = .750, \lambda_{21} = .853$??

- Model is **under-identified** if there are more unknown parameters than item variances, covariances, and means with which to estimate them
- Model **cannot be estimated** because there are an infinite number of different parameter estimates that would result in the same (and perfect) fit
- Example: $x + y = 7$??



Total possible **DF** = unique pieces of data = 9

0 factor variances

0 factor means

3 item loadings

3 item intercepts

3 error variances

OR

1 factor variance

1 factor mean

2 item loadings

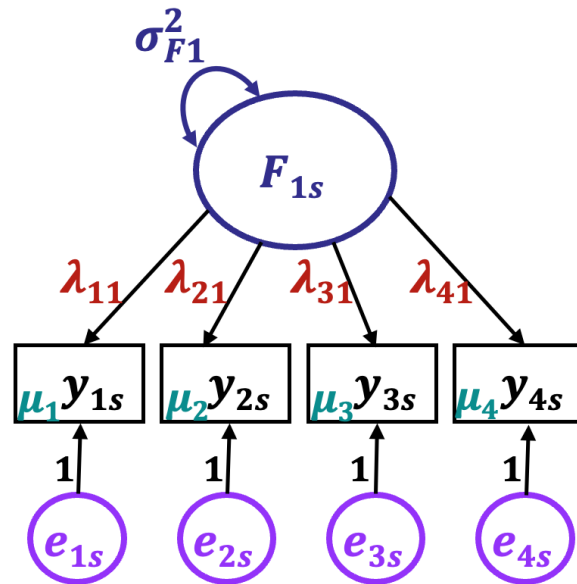
2 item intercepts

3 error variances

$$\text{Model DF} = 9 - 9 = 0$$

Not really a model—more like a description

- Model is **over-identified** if there are fewer unknown parameters than item variances, covariances, and means with which to estimate them
 - The model is **estimable**, so the parameter estimates have a unique solution
 - But now the parameters will NOT perfectly reproduce the observed matrix
→ if **Model DF > 0**, we can now test model fit!



Total possible **DF** = unique pieces of data = **14**

0 factor variances

0 factor means

4 item loadings

4 item intercepts

4 error variances

OR

1 factor variance

1 factor mean

3 item loadings

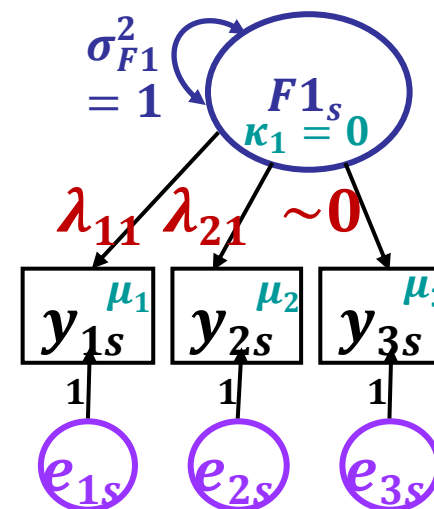
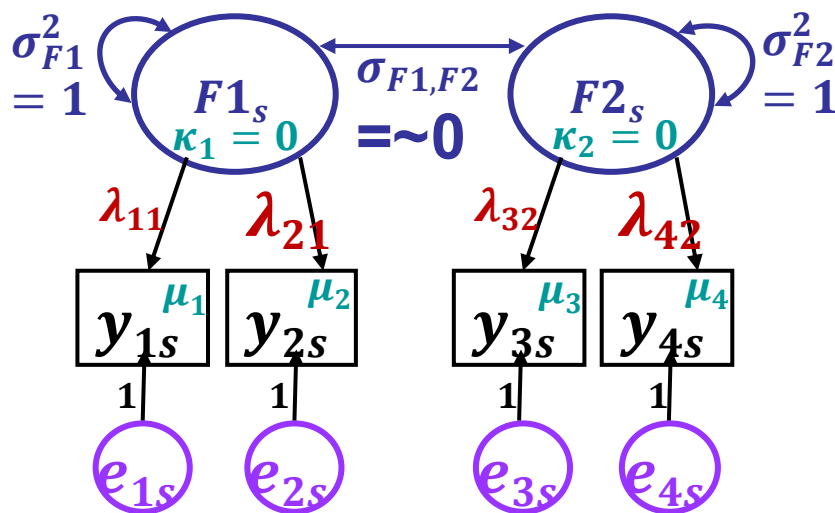
3 item intercepts

4 error variances

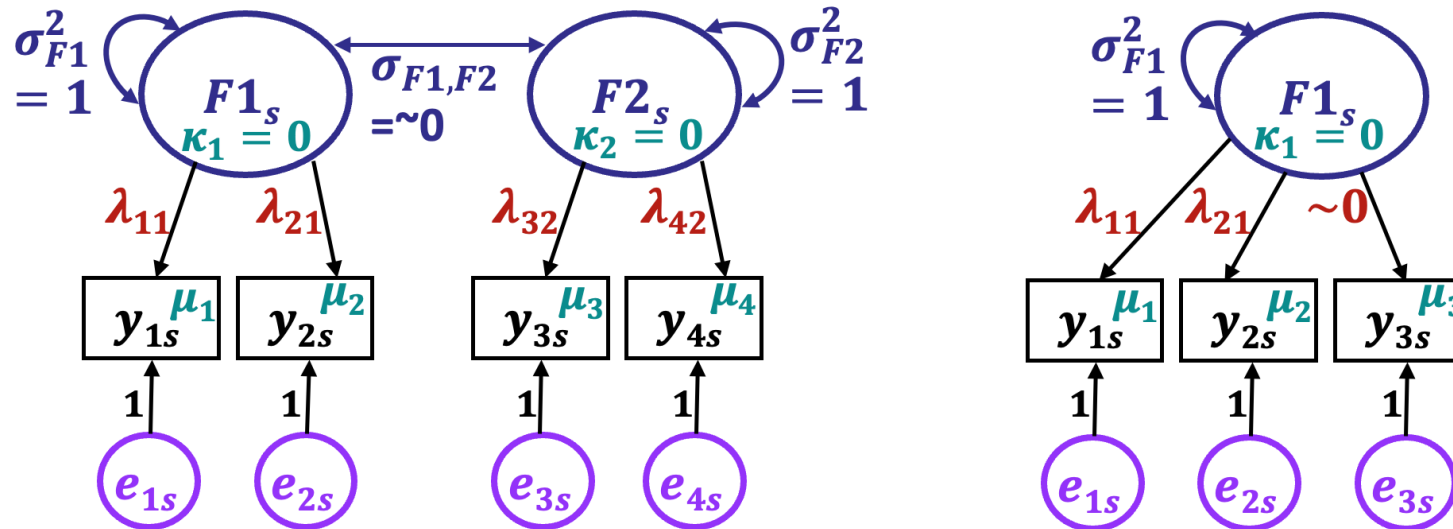
$$\text{Model DF} = 14 - 12 = 2$$

Model fit: Did we do a “good enough” job reproducing the item covariance matrix with 2 fewer parameters than it was possible to use?

- Did your model blow up (errors instead of output)? Double-check:
 - Part 1: Make sure each factor has a scale: a mean and a variance
 - Part 2: Make sure you aren't estimating more parameters than you have DF
- Sometimes you can set up your model correctly and it will STILL blow up because of **empirical under-identification**
 - It's not you; **it's your data**—here are two examples of when these models should have been identified, but weren't because of an unexpected 0 relationship



- Did your model blow up (errors instead of output)? Double-check:
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```
#model 2 syntax
model02.syntax = "

# measurement model (loading specification)
happiness =~ X1 + X2 + X3 + X4 + X5

# item intercepts
X1 ~ 1
X2 ~ 1
X3 ~ 1
X4 ~ 1
X5 ~ 1

# item residual (unique) variances
X1 ~~ X1
X2 ~~ X2
X3 ~~ X3
X4 ~~ X4
X5 ~~ X5

# happiness mean (fixed to zero for identification)
happiness ~ 0*1

# happiness variance (estimated as loadings use marker item approach)
happiness ~~ happiness

"
```

Step 1: Investigate Model Fit

```
> summary(model02.fit, fit.measures=TRUE, rsquare=TRUE, standardized=TRUE)
lavaan 0.6-19 ended normally after 55 iterations
```

Estimator	ML
Optimization method	NLMINB
Number of model parameters	15
Number of observations	150
Number of missing patterns	1

Model Test User Model:

	Standard	Scaled
Test Statistic	1.901	2.010
Degrees of freedom	5	5
P-value (Chi-square)	0.863	0.848
Scaling correction factor		0.946
Yuan-Bentler correction (Mplus variant)		

Model Test Baseline Model:

Test statistic	13.377	13.697
Degrees of freedom	10	10
P-value	0.203	0.187
Scaling correction factor		0.977

User Model versus Baseline Model:

Comparative Fit Index (CFI)	1.000	1.000
Tucker-Lewis Index (TLI)	2.836	2.618
Robust Comparative Fit Index (CFI)		1.000
Robust Tucker-Lewis Index (TLI)		2.719

Loglikelihood and Information Criteria:

Loglikelihood user model (H0)	-1086.457	-1086.457
Scaling correction factor for the MLR correction		0.963
Loglikelihood unrestricted model (H1)	-1085.507	-1085.507
Scaling correction factor for the MLR correction		0.959
Akaike (AIC)	2202.915	2202.915
Bayesian (BIC)	2248.074	2248.074
Sample-size adjusted Bayesian (SABIC)	2200.602	2200.602

Root Mean Square Error of Approximation:

RMSEA	0.000	0.000
90 Percent confidence interval - lower	0.000	0.000
90 Percent confidence interval - upper	0.060	0.067
P-value H_0: RMSEA <= 0.050	0.932	0.916
P-value H_0: RMSEA >= 0.080	0.022	0.029
Robust RMSEA		0.000
90 Percent confidence interval - lower		0.000
90 Percent confidence interval - upper		0.060
P-value H_0: Robust RMSEA <= 0.050		0.930
P-value H_0: Robust RMSEA >= 0.080		0.022

Standardized Root Mean Square Residual:

SRMR	0.024	0.024
------	-------	-------

- The model in our example fit (no surprise as it was with simulated data built from the same model)...but if it didn't, we could use the same tools we had in path analysis
- Residual covariances (for where fit is poor):

```
> resid(object = model02.fit, type = "normalized")
$type
[1] "normalized"

$cov
      X1      X2      X3      X4      X5
X1  0.000
X2 -0.841  0.000
X3  0.191  0.841  0.000
X4 -0.324 -0.403  0.128  0.000
X5  0.029  0.049 -0.062  0.027  0.000

$mean
X1 X2 X3 X4 X5
0  0  0  0  0
```

- Modification indices (for how we could improve fit):

```
> modificationindices(object = model02.fit, sort. = TRUE)
  lhs op rhs   mi   epc sepc.lv sepc.all sepc.nox
18 X1 ~~ X2 0.815 -0.086 -0.086 -0.075 -0.075
22 X2 ~~ X3 0.759  0.078  0.078  0.073  0.073
26 X3 ~~ X5 0.378 -0.211 -0.211 -0.319 -0.319
23 X2 ~~ X4 0.154 -0.035 -0.035 -0.033 -0.033
20 X1 ~~ X4 0.133 -0.039 -0.039 -0.037 -0.037
19 X1 ~~ X3 0.076  0.029  0.029  0.027  0.027
27 X4 ~~ X5 0.074  0.099  0.099  0.150  0.150
21 X1 ~~ X5 0.068  0.090  0.090  0.127  0.127
24 X2 ~~ X5 0.059  0.045  0.045  0.063  0.063
25 X3 ~~ X4 0.030  0.018  0.018  0.018  0.018
```

Step 2: Evaluate Model Parameters

Latent Variables:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
happiness =~						
X1	1.000				0.220	0.202
X2	0.175	0.588	0.298	0.766	0.039	0.036
X3	0.978	0.738	1.326	0.185	0.215	0.211
X4	1.011	0.800	1.263	0.207	0.222	0.219
X5	3.368	3.863	0.872	0.383	0.740	0.745

Intercepts:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.X1	3.027	0.089	34.023	0.000	3.027	2.778
.X2	2.931	0.088	33.389	0.000	2.931	2.726
.X3	2.855	0.083	34.269	0.000	2.855	2.798
.X4	2.939	0.083	35.471	0.000	2.939	2.896
.X5	2.987	0.081	36.832	0.000	2.987	3.007
happiness	0.000				0.000	0.000

Variances:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
.X1	1.139	0.136	8.380	0.000	1.139	0.959
.X2	1.154	0.129	8.948	0.000	1.154	0.999
.X3	0.995	0.120	8.288	0.000	0.995	0.956
.X4	0.981	0.113	8.711	0.000	0.981	0.952
.X5	0.440	0.597	0.736	0.462	0.440	0.445
happiness	0.048	0.067	0.721	0.471	1.000	1.000

R-Square:

	Estimate
X1	0.041
X2	0.001
X3	0.044
X4	0.048
X5	0.555

MOTIVATING STRUCTURAL EQUATION MODELING

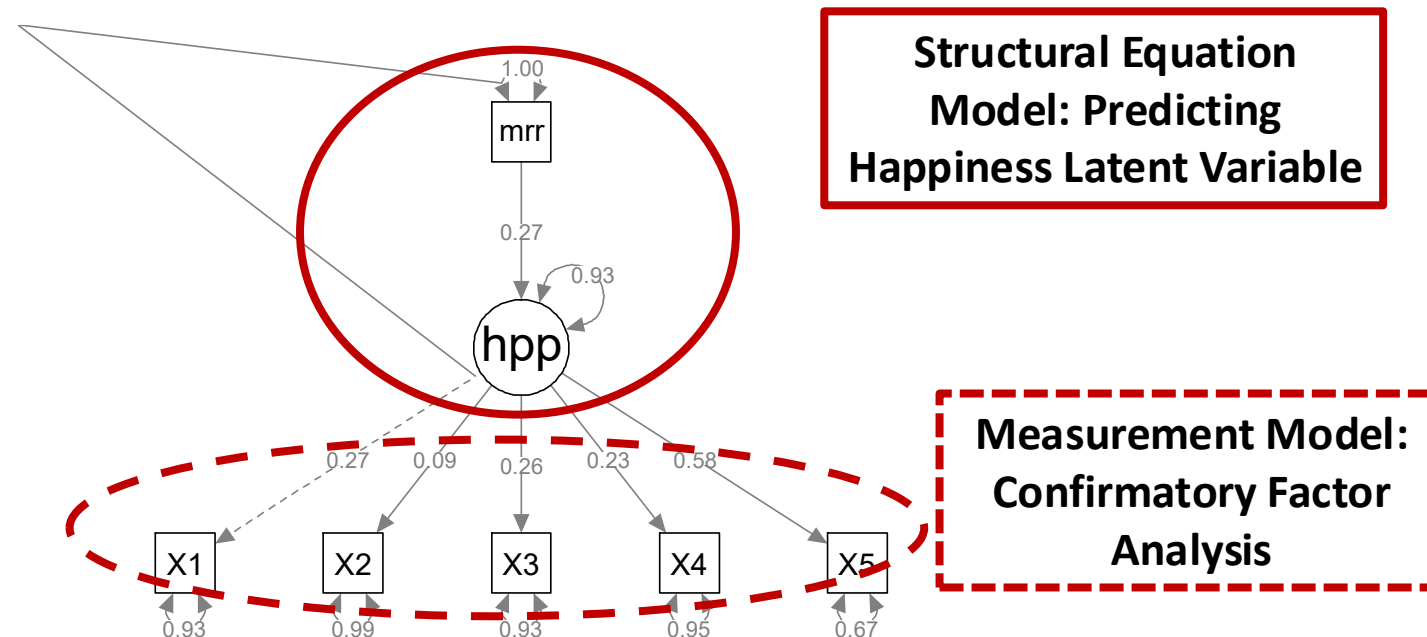
- When the GLM (regression/ANOVA/all other forms) are used with variables that are not measured perfectly, several deleterious things can happen:
 - The effects may be biased
 - Depends on variable type (i.e., how continuous are your measures)
 - The standard errors may be biased
 - Happens any time
 - Therefore...hypothesis tests may not be accurate
- Basically, any conclusions you make can be drawn into question when you have not-100%-reliable variables used
 - The issue is with just how reliable is reliable enough

- Structural Equation Modeling seeks to determine the relationship between
 - Latent constructs only
 - Latent our example: Happiness
 - Latent and observed constructs
 - From our example: how does marital status factor into the model?
 - Complex relationships between latent constructs and observed variables
 - More variables needed for our example
 - Includes mediation models
- SEM is a generalization of linear modeling using observed and latent (sometimes called random) variables

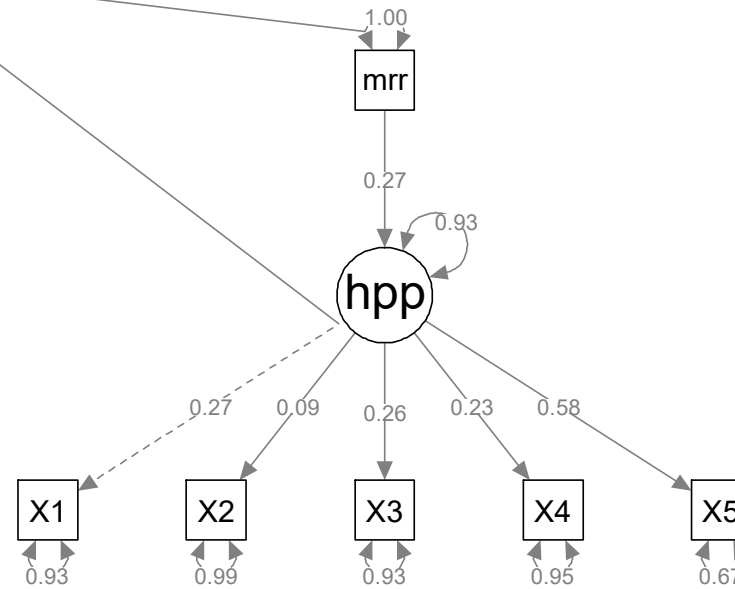
- A common way of depicting SEMs is with a path diagram → a pictorial representation of the statistical model
 - Observed variables: Squares
 - Latent variables: Circles
 - Direct effects: Arrows with one head
 - Indirect effects: Arrows with two heads
- From our previous GLM example
- Here MRR is marital status and hp_ is the happiness sum score



- Variables that are measured with error are represented as “latent” constructs in SEM
 - The latent variables are estimated directly by the model
 - Any equations involving latent variables are estimated simultaneously
- A more accurate depiction of our example:



SEM is often called
Path Analysis with
Latent Variables



$$\begin{aligned}X_{p1} &= \mu_1 + \lambda_1 HPP_p + e_{p1} \\X_{p2} &= \mu_2 + \lambda_2 HPP_p + e_{p2} \\X_{p3} &= \mu_3 + \lambda_3 HPP_p + e_{p3} \\X_{p4} &= \mu_4 + \lambda_4 HPP_p + e_{p4} \\X_{p5} &= \mu_5 + \lambda_5 HPP_p + e_{p5} \\HPP_p &= \beta_0 + \beta_1 Married_p + e_p^{HPP}\end{aligned}$$

- Recall the standardized coefficient from our GLM analysis:

Regressions:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
happiness_sumscore ~						
married	1.077	0.415	2.595	0.009	1.077	0.207

- Now here is the same coefficient from the SEM analysis:

Regressions:

	Estimate	Std.Err	z-value	P(> z)	Std.lv	Std.all
happiness ~						
married	0.159	0.168	0.945	0.344	0.546	0.272

- The GLM β_1 is lower—Look at std.all (biased due to unreliable Y)
- The GLM β_1 Wald test (z-value column) is smaller showing how the standard error is reduced more than the estimate
- The p-value is non-significant → the conclusion changes

- Statistical distributions are what drive the process
 - Each distribution is described by a set of parameters
 - Think of the normal distribution (mean and variance)

- Each of the lines represents model parameters
 - The statistical distribution of the boxes and circles are described by the model parameters

- Model parameters provide constraints to the statistical distribution parameters
 - Reduce complexity of model
 - Provide for meaningful inference

- A model is bound by distributions assumed and, hence, the number of possible parameters
 - We will learn statistics and path models
 - Both are needed to be good at SEM

WRAPPING UP

- Today was an introduction to measurement and structural equation models
- Measurement models are statistical models that provide precise information about the trait being studied
 - Much more accurate and informative than just adding item scores together
 - More advanced than exploratory factor analysis
- Structural equation models combine latent and observed variables simultaneously
 - Any time you have variables with error, you should be using these methods